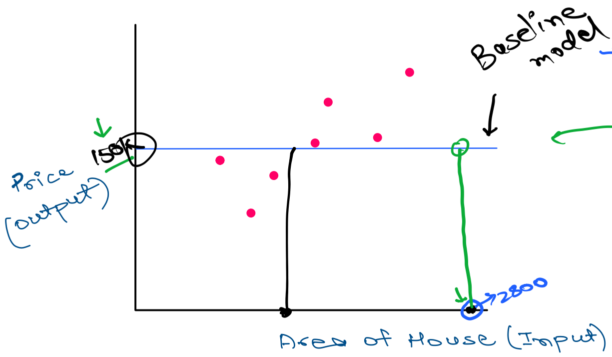
Linear Regression Rough Work 6 June

Saturday, 6 June 2026 19:00

Regression

Linear Regression



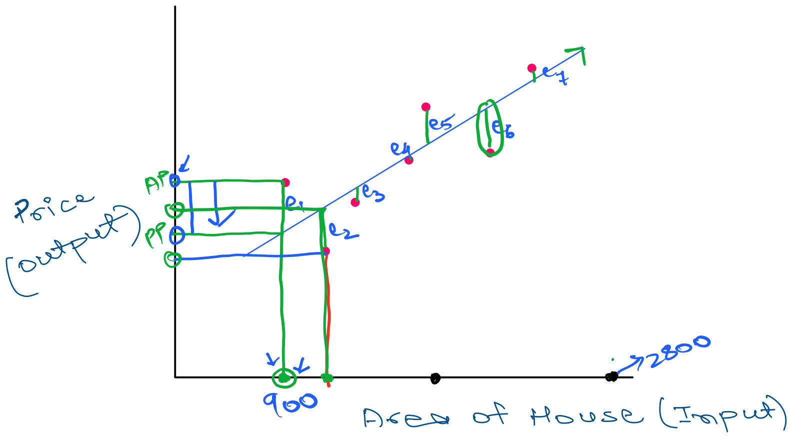
	input ↓ Area	o/p ↓ Price
0	1800	$90k
1	2500	$ 145k
2	4200	$ 280k
3	2100	$ 140k
4	–	–
5	–	–
6	–	–
7	2800	?

1300

Average of all the house prices

$150K

Smart solution → using Linear Regression Algo



Regression Line

↓

ML Model

Error = Actual Price − Pred Price

120000 − 105000

15000

$(15k)^2 + (8k)^2$

10k house/rows

	Area	Price	Pred Price
0	900	$120k	$ 105k
	1000	$100k	$108k

distance

Modulus
absolute
$|-10| = 10$

Total Error made by a line

$$TE = e_1 + e_2 + e_3 + e_4 + e_5 + e_6 + e_7$$

$$\downarrow$$

$$TE = e_1^2 + e_2^2 + e_3^2 + e_4^2 + e_5^2 + \cdots$$

Sum of squared errors (SSE)

$$\downarrow$$

Another prob: Explosion of values

$$\downarrow$$

$$MSE = \frac{SSE}{n} = \frac{e_1^2 + e_2^2 + e_3^2 + e_4^2 + \cdots}{n}$$

Mean squared error.

$$e_1 \quad e_2 \quad e_3 \quad e_4 \quad \cdots$$

$$\downarrow$$

$$✗\ TE = e_1 + e_2 + e_3 + \cdots$$

$$\downarrow$$

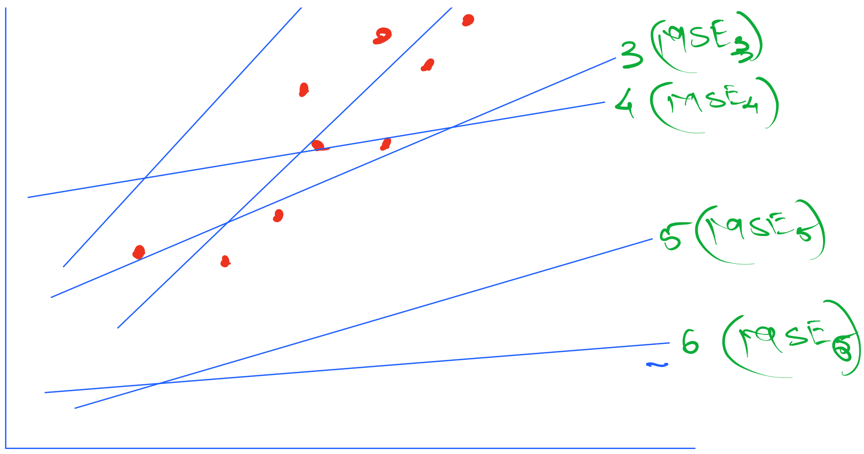
$$SSE = e_1^2 + e_2^2 + e_3^2 + \cdots$$

$$\downarrow$$

$$✓\ MSE = \frac{SSE}{n} = \frac{e_1^2 + e_2^2 + e_3^2 + \cdots}{n}$$

1 (MSE_1)

2 (MSE_2)



MSE_1 vs MSE_2 vs MSE_3 vs MSE_4 vs MSE_5 vs MSE

↓

will be lowest here,

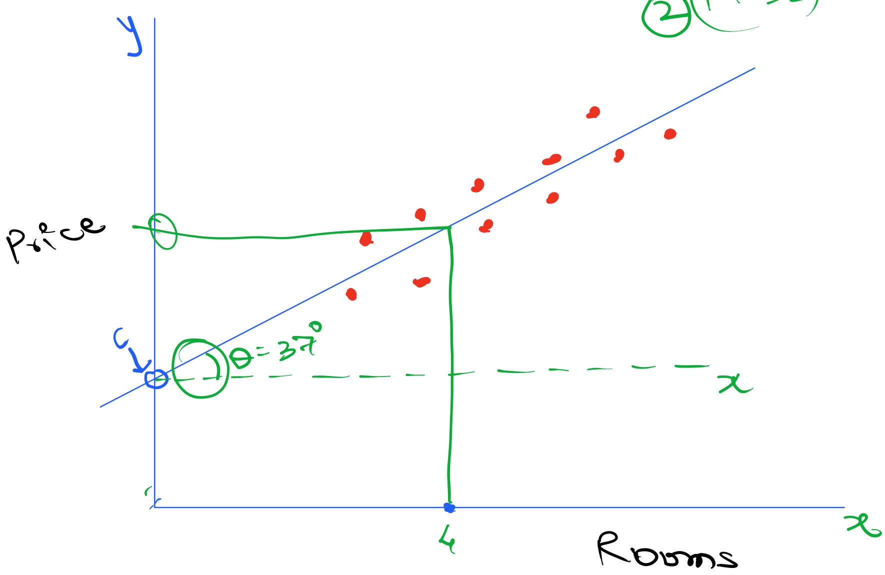
which means

Line 2 is most accurately trained Regression line

2 (MSE_2) → Best Fit Regression Line

↓

ML Model



$m = \tan\theta$

$m = \tan 37^\circ$

=

We need to represent our ML Model or Best Fit

Line as an equation:

$$y = mx + c$$

$$\boxed{\text{Price} = W * \text{Rooms} + c}$$

2 lines

Rooms: 4

$$\boxed{\text{Price} = 0.87 * \text{Rooms} + 0.81}$$

$$= 0.87 * 4 + 0.81$$

$$=$$

1. Simple Linear Regression
2. Multiple Linear regression

I/p cols, feature, ind var. | Target, output, dep var.

x_1 Area	x_2 Rooms	x_3 Floors	x_4 Age	y Price

$$\text{Price} = w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4 + c$$

$$= w_1 * \text{Area} + w_2 * \text{Rooms} + w_3 * \text{Floors} + w_4 * \text{Age} + c$$

Train a ML Model

Data (Training)

↓

ML Algorithm

↓

$\boxed{\text{ML Model}}$

No